Centre National de la Recherche Scientifique

Complete list of publications

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1 Journal articles under review

A65 Giacomo Benvenuti, Sandrine Chemla, Arjan Boonman, Laurent U Perrinet, Guillaume S Masson and Frédéric Y Chavane. 'Anticipatory Responses along Motion Trajectories in Awake Monkey Area V1'. en. In: *bioRxiv* (2020), p. 2020.03.26.010017. DOI: 10.1101/2020.03.26.010017. URL: https://www.biorxiv.org/content/10.1101/2020.03.26.010017

A64 Laurent U Perrinet. 'Working Memory with Polychronous Chains'. In: *arXiv preprint arXiv:2604.14096* (2026). DOI: 10.48550/arXiv.2604.14096. URL: https://laurentperrinet.github.io/publication/perrinet-26

A63 Matthis Dallain, Laurent Rodriguez, Laurent U Perrinet and Benoît Miramond. 'A Saccade-Inspired Approach to Image Classification Using Visiontransformer Attention Maps'. working paper. 10th Mar. 2026. DOI: 10.48550/arXiv.2603.09613. arXiv: 2603.09613 [cs]. URL: http://arxiv.org/abs/2603.09613 (visited on 11/03/2026). Pre-published

2 Peer-reviewed international journal articles

2026

A62 Kevin Mairot, Isabelle Meunier, Béatrice Bocquet, Natacha Stolowy, Prithvi Ramtohul, Xavier Zanlonghi, Danièle Denis, Thierry David, Frederic Matonti and Laurent Udo Perrinet. 'Deep Learning for Diagnosis of Choroideremia and USH2A-Associated RodCone Dystrophy Using Macular Optical Coherence Tomography Volumes'. In: *Ophthalmology Science* 6.10 (Oct. 2026), p. 101378. ISSN: 2666-9145. DOI: 10.1016/j.xops.2026.101378. URL: https://doi.org/10.1016/j.xops.2026.101378 (visited on 27/08/2026)

A61 Thomas Kronland-Martinet, Stéphane Viollet and Laurent U Perrinet. 'Detection of spiking motifs of arbitrary length in neural activity using bounded synaptic delays'. In: *Biological Cybernetics* 120.5–6 (17th Aug. 2026), p. 27. ISSN: 1432-0770. DOI: 10.1007/s00422-026-01051-7

A60 Jean-Nicolas Jérémie, Emmanuel Daucé and Laurent U. Perrinet. 'Foveated Retinotopy Improves Classification and Localization in Convolutional Neural Networks'. In: *Vision* 10.2 (June 2026), p. 17. ISSN: 2411-5150. DOI: 10.3390/vision10020017. URL: https://www.mdpi.com/2411-5150/10/2/17

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A59 Andrew Isaac Meso, Jonathan Vacher, Nikos Gekas, Pascal Mamassian, Laurent U Perrinet and Guillaume S Masson. 'DynTex: A Real-Time Generative Model of Dynamic Naturalistic Luminance Textures'. In: *Journal of Vision* 25.11 (Sept. 2025), p. 2. ISSN: 1534-7362. DOI: 10.1167/jov.25.11.2. URL: https://doi.org/10.1167/jov.25.11.2 (visited on 03/09/2025)

A58 Skye Gunasekaran, Assel Kembay, Hugo Ladret, Rui-Jie Zhu, Laurent U Perrinet, Omid Kavehei and Jason Eshraghian. 'A Predictive Approach to Enhance Time-Series Forecasting'. In: *Nature Communications* (19th Oct. 2025). DOI: 10.1038/s41467-025-63786-4. arXiv: 2410.15217. URL: https://www.nature.com/articles/s41467-025-63786-4

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A57 Antoine Grimaldi, Victor Boutin, Sio-Hoi Ieng, Ryad Benosman and Laurent U Perrinet. 'A Robust Event-Driven Approach to Always-on Object Recognition'. In: *Neural Networks* 178 (1st Oct. 2024), p. 106415. DOI: 10.1016/j.neunet.2024.106415. URL: https://laurentperrinet.github.io/publication/grimaldi-24/

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A55 Ilias Rentzeperis, Luca Calatroni, Laurent U Perrinet and Dario Prandi. 'Beyond ℓ_1 sparse coding in V1'. In: *PLOS Computational Biology* (12th Sept. 2023). DOI: 10.1371/journal.pcbi.1011459. URL: https://laurentperrinet.github.io/publication/rentzeperis-23

A54 Amélie Gruel, Dalia Hareb, Antoine Grimaldi, Jean Martinet, Laurent U Perrinet, Bernabé Linares-Barranco and Teresa Serrano-Gotarredona. 'Stakes of Neuromorphic Foveation: a promising future for embedded event cameras'. In: *Biological Cybernetics* (2023). DOI: 10.1007/s00422-023-00974-9

A53 Antoine Grimaldi and Laurent U Perrinet. 'Learning heterogeneous delays in a layer of spiking neurons for fast motion detection'. In: *Biological Cybernetics* (11th Sept. 2023). DOI: 10.1007/s00422-023-00975-8. URL: https://laurentperrinet.github.io/publication/grimaldi-23-bc/

A52 Hugo Ladret, Nelson Cortes, Lamyae Ikan, Frédéric Chavane, Christian Casanova and Laurent U Perrinet. 'Cortical recurrence supports resilience to sensory variance in the primary visual cortex'. In: *Nature Communications Biology* (6th June 2023). DOI: 10.1038/s42003-023-05042-3. URL: https://www.nature.com/articles/s42003-023-05042-3

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A49 Victor Boutin, Angelo Franciosini, Frédéric Y Chavane and Laurent U Perrinet. 'Pooling in a predictive model of V1 explains functional and structural diversity across species'. In: *PLoS Computational Biology* (18th July 2022). DOI: 10.1371/journal.pcbi.1010270. URL: https://laurentperrinet.github.io/publication/franciosini-21

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A44 Emmanuel Daucé, Pierre Albigès and Laurent U Perrinet. 'A dual foveal-peripheral visual processing model implements efficient saccade selection'. In: *Journal of Vision* 20.8 (5th June 2020), pp. 22–22. DOI: 10.1167/jov.20.8.22. URL: https://laurentperrinet.github.io/publication/dauce-20/

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A41 Cesar U Ravello, Laurent U Perrinet, Maria-José Escobar and Adrián G Palacios. 'Speed-Selectivity in Retinal Ganglion Cells is Sharpened by Broad Spatial Frequency, Naturalistic Stimuli'. In: *Scientific Reports* 9.1 (24th Jan. 2019). DOI: 10.1038/s41598-018-36861-8. URL: https://doi.org/10.1038%2Fs41598-018-36861-8

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A37 Mina A Khoei, Guillaume S Masson and Laurent U Perrinet. 'The flash-lag effect as a motion-based predictive shift'. In: *PLoS Computational Biology* 13.1 (26th Jan. 2017), e1005068. DOI: 10.1371/journal.pcbi.1005068. URL: https://laurentperrinet.github.io/publication/khoei-masson-perrinet-17/

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A36 Wahiba Taouali, Giacomo Benvenuti, Pascal Wallisch, Frédéric Y Chavane and Laurent U Perrinet. 'Testing the odds of inherent vs. observed overdispersion in neural spike counts'. In: *Journal of Neurophysiology* 115.1 (22nd Jan. 2016), pp. 434–444. ISSN: 1522-1598. DOI: 10.1152/jn.00194.2015. URL: http://www.ncbi.nlm.nih.gov/pubmed/26445864

A35 Jonathan Vacher, Andrew Isaac Meso, Laurent U Perrinet and Gabriel Peyré. 'Biologically Inspired Dynamic Textures for Probing Motion Perception'. In: *Advances in Neural Information Processing Systems* 28 (2015), pp. 1918–1926. URL: http://papers.nips.cc/paper/5769-biologically-inspired-dynamic-textures-for-probing-motion-perception.pdf

A34 Laurent U Perrinet and James A Bednar. 'Edge co-occurrences can account for rapid categorization of natural versus animal images'. In: *Scientific Reports* 5 (2015), p. 11400. DOI: 10.1038/srep11400. URL: http://www.nature.com/articles/srep11400

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A33 Laurent U Perrinet, Rick A Adams and Karl Friston. 'Active inference, eye movements and oculomotor delays'. In: *Biological Cybernetics* 108.6 (16th Dec. 2014), pp. 777–801. ISSN: 1432-0770. DOI: 10.1007/s00422-014-0620-8

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A32 Mina A Khoei, Guillaume S Masson and Laurent U Perrinet. 'Motion-based prediction explains the role of tracking in motion extrapolation'. In: *Journal of Physiology-Paris* 107.5 (1st Nov. 2013), pp. 409–420. ISSN: 0928-4257. DOI: 10.1016/j.jphysparis.2013.08.001. URL: https://laurentperrinet.github.io/publication/khoei-13-jpp/

A31 Bernhard A Kaplan, Anders Lansner, Guillaume S Masson and Laurent U Perrinet. 'Anisotropic connectivity implements motion-based prediction in a spiking neural network'. In: *Frontiers in Computational Neuroscience* 7.112 (17th Sept. 2013). DOI: 10.3389/fncom.2013.00112. URL: https://laurentperrinet.github.io/publication/kaplan-13

A30 Rodrigo Nava, J Victor Marcos, Boris Escalante-Ramirez, Gabriel Cristóbal, Laurent U Perrinet and Raúl S J Estépar. 'Advances in Texture Analysis for Emphysema Classification'. In: *Lecture Notes in Computer Science* 8259 (2013). Ed. by David Hutchison et al., pp. 214–221. ISSN: 1611-3349. DOI: 10.1007/978-3-642-41827-3_27

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A22 Jérôme Fleuriet, Sandrine Hugues, Laurent U Perrinet and Laurent Goffart. 'Saccadic foveation of a moving visual target in the rhesus monkey'. In: *Journal of Neurophysiology* 105.2 (1st Feb. 2011), pp. 883–895. ISSN: 1522-1598. DOI: 10.1152/jn.00622.2010. URL: http://dx.doi.org/10.1152/jn.00622.2010

A21 Amarender Bogadhi, Anna Montagnini, Pascal Mamassian, Laurent U Perrinet and Guillaume S Masson. 'Pursuing motion illusions: a realistic oculomotor framework for Bayesian inference'. In: *Vision research* 51.8 (22nd Apr. 2011), pp. 867–880. ISSN: 1878-5646. DOI: 10.1016/j.visres.2010.10.021. URL: http://dx.doi.org/10.1016/j.visres.2010.10.021

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3 Peer-reviewed book chapters

C6 Laurent U Perrinet. 'From the retina to action: Dynamics of predictive processing in the visual system'. In: *The Philosophy and Science of Predictive Processing*. Ed. by Dina Mendonça, Manuel Curado and Steven S Gouveia. Bloomsbury, 2020, pp. 85–104. DOI: 10.5040/9781350099784.ch-005. URL: https://laurentperrinet.github.io/Perrinet20PredictiveProcessing_manubot/v/latest/index.html

C5 Anna Montagnini, Laurent U Perrinet and Guillaume S Masson. 'Visual motion processing and human tracking behavior'. In: *Biologically Inspired Computer Vision*. Ed. by Gabriel Cristóbal, Laurent U Perrinet and Matthias S Keil. Wiley-VCH Verlag GmbH and Co. KGaA, 1st Nov. 2015. Chap. 12. DOI: 10.1002/9783527680863.ch12. URL: https://laurentperrinet.github.io/publication/montagnini-15-bicv/

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C2 Bruno Cessac, Emmanuel Daucé, Laurent U Perrinet and Manuel Samuelides. 'Introduction to Topics in Dynamical Neural Networks: From Large Scale Neural Networks to Motor Control and Vision'. In: *Topics in Dynamical Neural Networks: From Large Scale Neural Networks to Motor Control and Vision*. Vol. 142. The European Physical Journal Special Topics 1. Springer Verlag, 1st Mar. 2007, pp. 1–5. DOI: 10.1140/epjst/e2007-00057-3. URL: http://www.springerlink.com/index/10.1140/epjst/e2007-00057-3

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4 Theses, projects and books

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